

ECE 3210 Midterm 1

Week of: October 6, 2025

Student's name: _____

Instructor:

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You have 2.5 hours for 5 problems.

- Show enough (neat) work in the clear spaces on this exam to convince us that you derived, not guessed, your answers.
- Put your final answers in the boxes at the bottom of the page.

You may use one US letter-style size page of notes, front and back.

Problem	Score	Possible Points
1		20
2		20
3		20
4		20
5		20
Total score		100

1 Short answer

- (a) What is the verb form of “convolution”?

- (b) Consider the following systems. Determine if they are stable, marginally stable, or unstable.

(i) $(D^2 + 1)(D - 3)y(t) = (D + 1)x(t)$

Circle one:	stable	marginally stable	unstable
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(ii) $h(t) = t^t [u(t) - u(t - 10)]$.

Circle one:	stable	marginally stable	unstable
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(iii) $(D^2 - 4)y(t) = (D^2 + 1)x(t)$

Circle one:	stable	marginally stable	unstable
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- (c) Consider the following systems. Determine if they are linear and/or time-invariant. No need to write out a formal proof but feel free to use the space to work through the problem if needed.

(i) $y(t) = x(t) \cdot \frac{d[x(t)]}{dt}$

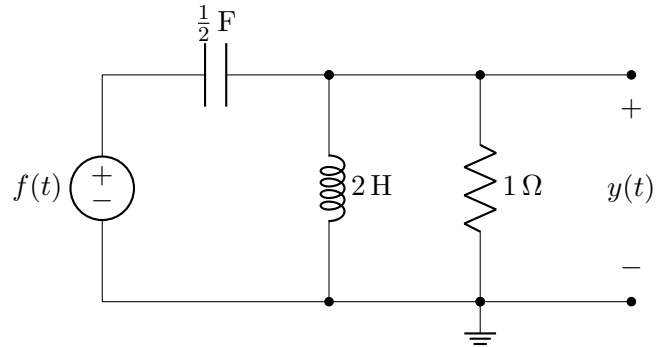
Circle one:	linear	non-linear
Circle one:	time-invariant	time-variant

(ii) $y(t) = x(t) + \cos(t)$

Circle one:	linear	non-linear
Circle one:	time-invariant	time-variant

2 Impulse response

Consider the circuit below.



- (a) Derive the ODE relating the input $f(t)$ and the output $y(t)$. In your final answer, don't worry about including the units, just write the ODE in the form we have seen in class (you can use the D operator, the $\frac{d}{dt}$ operator, etc.).

- (b) What is the order of this system?

Order =

- (c) What is b_n ?

$b_n =$

- (d) Assuming initial conditions $y_n(0) = 0$ and $y'_n(0) = 1$, find the zero-input response for this system $y_2(t)$.

$y_2(t) =$

- (e) Find the impulse response $h(t)$ for this system.

$h(t) =$

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3 Convolution

Perform the following convolutions. Simplify as much possible.

(a) $y(t) = u(t) * e^{-2t} (u(t) - u(t - 2))$

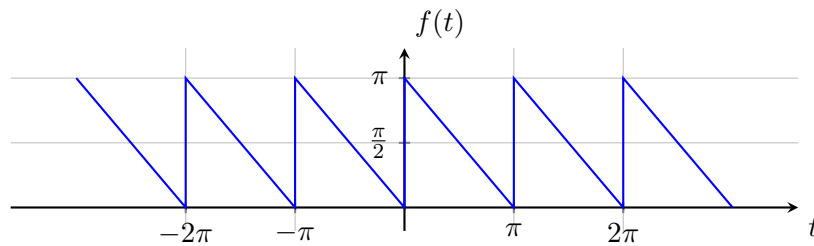
$$y(t) =$$

(b) $y(t) = \sin(\pi t) * \text{rect}(t)$

$y(t) =$

4 LTI system response to periodic input

- (a) Consider the periodic input to some system $f(t)$ shown below.



Determine the period T and fundamental frequency ω_0 for the function $f(t)$.

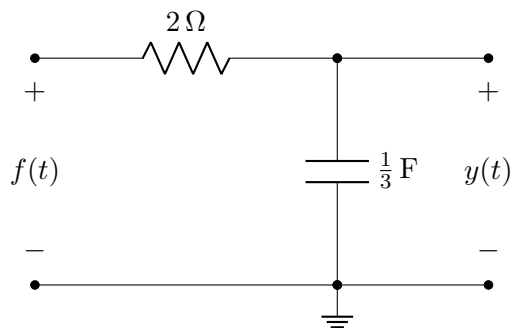
$T =$

$\omega_0 =$

- (b) Derive the *complex exponential* Fourier series for $f(t)$. *Hint: Be thoughtful about which period you want to integrate over.*

$f(t) =$

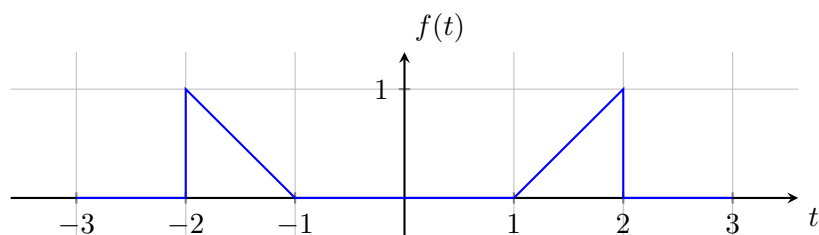
- (c) Given the input $f(t)$ used in parts (a) and (b) for the system with transfer function below, what is the output $y(t)$? If you couldn't solve part (b), you can just use “ D_n ” in your series expansion.



$y(t) =$

5 Fourier transforms

(a) Consider the following time-domain signal.



Determine the Fourier transform $F(\omega)$ for the signal $f(t)$ shown above.

$$F(\omega) =$$

(b) Evaluate the integral

$$I = \int_{-\infty}^{\infty} \text{sinc}(t) dt.$$

$I =$

(c) BONUS (+10 points if correct): Evaluate the integral

$$I = \int_{-\infty}^{\infty} \frac{2\text{sinc}(t)}{t^2 + 1} dt$$

$I =$